THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET A

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A constant force of 20 N displaces a body by 5 m in the direction of the force. The work done is: 1

(a) 200 J **(b)** 4 J **(c)** 100 J **(d)** 25 J

2. The kinetic energy of a body of mass 2 kg moving at 4 m s^{-1} is: 1

(a) 32 J **(b)** 8 J **(c)** 26 J **(d)** 16 J

3. The potential energy of a 2 kg body at a height of 5 m (g = 10 m s^{-2}) is: 1

(a) 100 J **(b)** 50 J **(c)** 10 J **(d)** 200 J

4. A machine does 600 J of work in 10 s. Its power is: 1

(a) 6000 W **(b)** 0.02 W **(c)** 60 W **(d)** 120 W

5. A machine lifts a load of 300 N by applying an effort of 100 N. Its mechanical advantage is: 1

(a) 6 **(b)** 0.33 **(c)** 4 **(d)** 3

6. On a lever the load is 60 N at 20 cm from the fulcrum. To balance it with the effort arm 40 cm, the effort needed is: 1

(a) 60 N **(b)** 80 N **(c)** 30 N **(d)** 15 N

7. A girl carrying a box walks horizontally. The work done by the force holding the box is: 1

(a) zero **(b)** negative **(c)** positive **(d)** mgh

8. As a body falls freely, its mechanical energy: 1

(a) decreases **(b)** becomes zero **(c)** increases **(d)** remains constant

9. A wheelbarrow is a lever with the: 1

(a) fulcrum between load and effort **(b)** effort between load and fulcrum

(c) no load **(d)** load between fulcrum and effort

10. Two bodies A and B of masses m and 4m have the same kinetic energy. The ratio of their speeds vA : vB is: 1

(a) 1 : 2 **(b)** 1 : 1 **(c)** 4 : 1 **(d)** 2 : 1

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Define work in the scientific sense. Give its SI unit. 2

12. Define 1 joule of work. 2

13. Under what conditions is the work done by a force zero? 2

14. A force of 40 N moves a trolley by 5 m in the direction of the force. Find the work done. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A girl lifts a dumbbell up and then lowers it. When is positive work done and when negative? 3

16. Explain why no work is done when you push hard against a wall. 3

17. A goalkeeper stops a ball with a force of 200 N, her hands moving back 15 cm. Find the work done on the ball. 3

OR

Derive the expression for kinetic energy. 3

18. The velocity of a vehicle is doubled. How does its kinetic energy change? Explain. 3

OR

A 0.2 kg cricket ball is bowled at 43 m s^{-1}. Find its kinetic energy. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Define work and joule.
(ii) When is work zero, positive or negative?
(iii) A force of 25 N moves a body 6 m; find the work. 5

OR

Case study: A girl lifts a 5 kg bag to 1 m, then three bags together to 1 m, and a single bag to 3 m. 5
(i) Compare the work in the three cases.
(ii) Find the work for one bag (g = 10).
(iii) Explain the relation between work, force and distance.

20. (i) State the work-energy theorem.
(ii) Derive the expression for kinetic energy.
(iii) Find the speed of a 15 000 kg aircraft stopped within 100 m by a force of 367 500 N. 5

OR

(i) Derive PE = mgh. 5
(ii) Describe the sand-bed activity to show that PE depends on height.
(iii) What is meant by conservation of mechanical energy?

END OF QUESTION PAPER — Total: 40 Marks

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